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Date: 1st February 2025

Batch: AI09
Course: Data Processing for AI

Section A: SQL Case Study- Employee Management System

Q1: Create all 5 tables in PostgreSQL and insert the above data

The screenshot shows the SQLiteonline.com web interface. On the left sidebar, the 'PostgreSQL' database is selected, and a table named 'employees' is highlighted. The main area displays the SQL query `SELECT * FROM employees;` and the resulting table data:

employee_id	name	manager_id	department_id
1	Alice	NULL	1
2	Bob	1	2
3	Charlie	1	3
4	David	2	2
5	Eve	3	3
6	Frank	2	2

Q2: Write an SQL query to list all employees along with their manager's name.

The screenshot shows the SQLiteonline.com web interface. The SQL query entered is:

```
1 -- Selecting employee names along with their respective manager names
2 SELECT e.name AS employee_name,
3        COALESCE(M.name, 'No Manager') AS manager_name -- If manager_id is NULL, display 'No Manager'
4 FROM employees e
5 LEFT JOIN employees M ON e.manager_id = M.employee_id; -- Joining employees table to get manager details
6
```

The result set shows the following data:

employee_name	manager_name
Alice	No Manager
Bob	Alice
Charlie	Alice
David	Bob
Eve	Charlie
Frank	Bob

Q3: Write an SQL query to find all employees who earn more than the average salary.

The screenshot shows the SQLiteonline.com web application. On the left sidebar, the 'PostgreSQL' database is selected, and the 'employees' table is highlighted. The main editor area contains the following SQL query:

```
1 -- Selecting employees whose salary is greater than the average salary
2 SELECT e.name, s.salary
3 FROM employees e
4 JOIN salaries s ON e.employee_id = s.employee_id
5 WHERE s.salary > (SELECT AVG(salary) FROM salaries); -- Subquery to get the average salary
6
```

The query results are displayed in a table with two columns: 'name' and 'salary'.

name	salary
Alice	90000
Bob	70000
Charlie	75000

Q4: Write an SQL query to list all employees along with the projects they are assigned to. If an employee is not assigned to any project, show NULL in the project column.

The screenshot shows the SQLiteonline.com web application. On the left sidebar, the 'PostgreSQL' database is selected, and the 'employee_projects' table is highlighted. The main editor area contains the following SQL query:

```
1 -- Fetching employees along with the projects they are assigned to
2 SELECT e.name AS employee_name, P.project_name
3 FROM employees e
4 LEFT JOIN employee_projects ep ON e.employee_id = ep.employee_id
5 LEFT JOIN projects P ON ep.project_id = P.project_id;
6
```

The query results are displayed in a table with two columns: 'employee_name' and 'project_name'.

employee_name	project_name
Bob	AI Development
Charlie	Sales Dashboard
David	AI Development
Eve	Sales Dashboard
Frank	AI Development
Alice	NULL

Q5: Write an SQL query to find all departments where the average salary is greater than 60,000.

The screenshot shows the SQLiteOnline.com web interface. On the left sidebar, the 'PostgreSQL' database is selected, and the 'employees' table is highlighted. The main editor contains the following SQL query:

```
1 -- Selecting departments where the average salary of employees is greater than 60,000
2 SELECT d.department_name
3 FROM employees e
4 JOIN salaries s ON e.employee_id = s.employee_id
5 JOIN departments d ON e.department_id = d.department_id
6 GROUP BY d.department_name
7 HAVING AVG(s.salary) > 60000; -- Filtering departments where average salary is greater than 60,000
8
```

The results pane on the right shows the output of the query:

department_name
Sales
HR

Q6: Write an SQL query to classify each employee into salary categories: "High Salary" if salary > 80,000, "Medium Salary" if salary is between 50,000 and 80,000, "Low Salary" if salary is below 50,000

The screenshot shows the SQLiteOnline.com web interface. On the left sidebar, the 'PostgreSQL' database is selected, and the 'employees' table is highlighted. The main editor contains the following SQL query:

```
1 -- Categorizing employees based on their salary
2 SELECT e.name, s.salary,
3        CASE
4            WHEN s.salary > 80000 THEN 'High Salary' -- Salary above 80,000
5            WHEN s.salary BETWEEN 50000 AND 80000 THEN 'Medium Salary' -- Salary between 50,000 and 80,000
6            ELSE 'Low Salary' -- Salary below 50,000
7        END AS salary_category
8 FROM employees e
9 JOIN salaries s ON e.employee_id = s.employee_id;
10
```

The results pane on the right shows the output of the query:

name	salary	salary_category
Alice	90000	High Salary
Bob	70000	Medium Salary
Charlie	75000	Medium Salary
David	50000	Medium Salary
Eve	48000	Low Salary
Frank	52000	Medium Salary

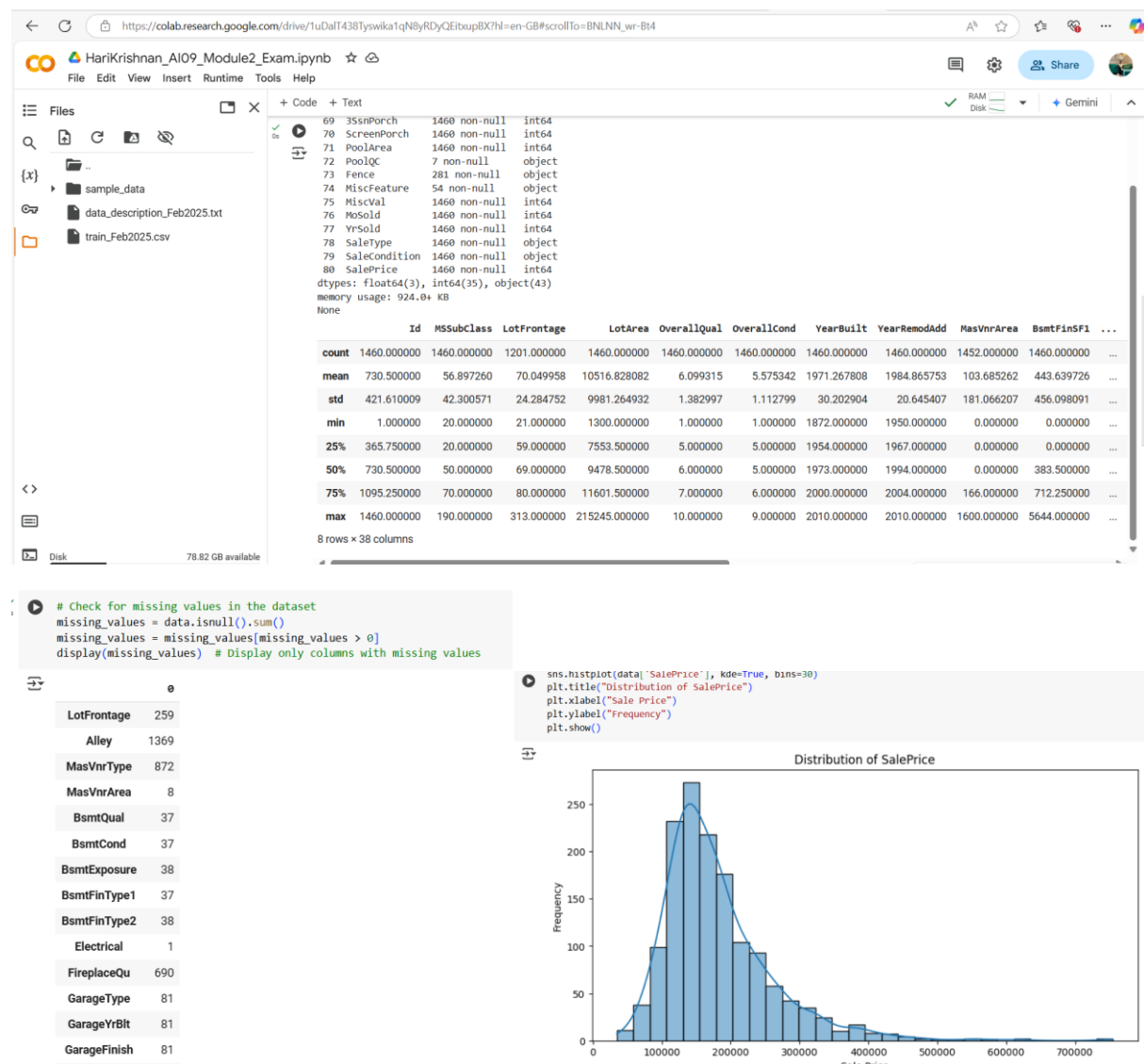
Section B: For the provided dataset (train_Feb2025.csv'), use the file data_description_Feb2025.txt' to understand each variable. The variable 'SalePrice' represents the property sale price in dollars and is the target variable. Based on your understanding of the data, answer the following questions.

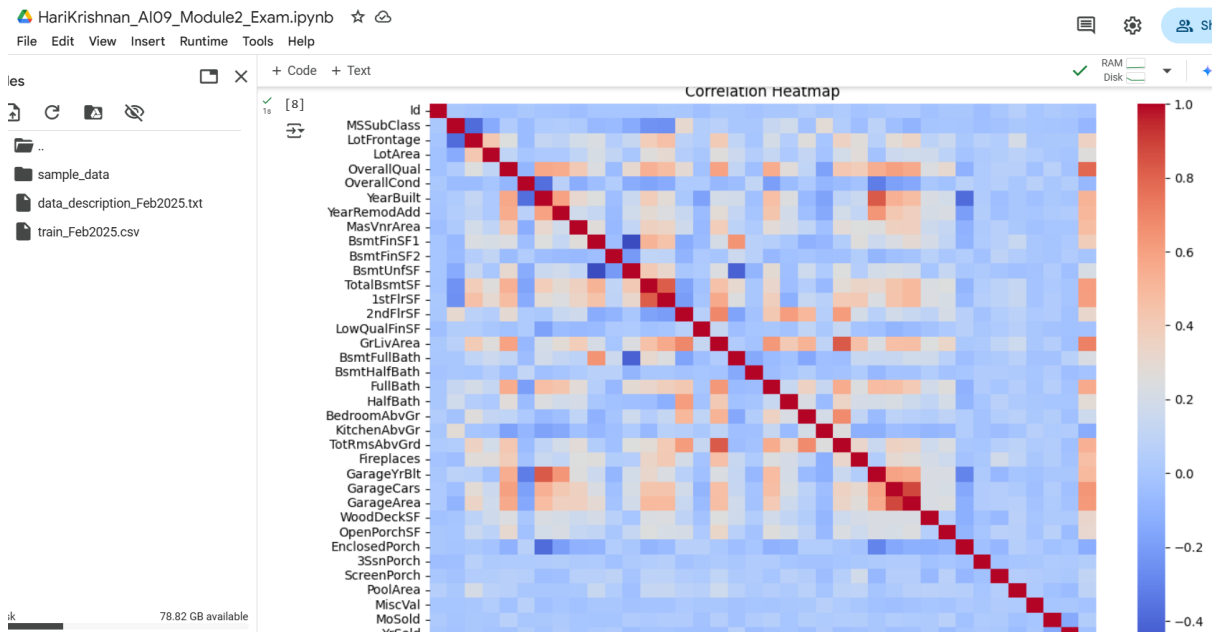
Submit your Jupyter notebook along with the output of your work in HTML format, including detailed explanations and insights.

Q7: Detailed EDA:

a. Conduct a comprehensive Exploratory Data Analysis (EDA) on the dataset. This should include visualizing distributions, exploring relationships between variables, and summarizing key findings.

b. Provide insights and interpretations that could be useful for building predictive models. You don't need to build any predictive models, just perform a detailed analysis and EDA of the data.





Q8: Dummy Coding:

a. Perform dummy coding on the categorical variables to incorporate them into a machine learning model. Explain the process and rationale.

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les

sample_data
data_description_Feb2025.txt
train_Feb2025.csv

78.82 GB available

```
[8]
[9]
[10] # Q8: Dummy Coding for Categorical Variables
      # Identifying categorical columns for encoding
      categorical_cols = data.select_dtypes(include=['object']).columns

      # Applying One-Hot Encoding to categorical variables
      dummy_data = pd.get_dummies(data, columns=categorical_cols, drop_first=True)

      display(dummy_data.head()) # Displaying the first few rows of the transformed dataset
```

	Id	MSSubClass	LotFrontage	LotArea	OverallQual	OverallCond	YearBuilt	YearRemodAdd	MasVnrArea	BsmtFinSF1	...	SaleType_ConLI	SaleTy
0	1	60	65.0	8450	7	5	2003	2003	196.0	706	...	False	
1	2	20	80.0	9600	6	8	1976	1976	0.0	978	...	False	
2	3	60	68.0	11250	7	5	2001	2002	162.0	486	...	False	
3	4	70	60.0	9550	7	5	1915	1970	0.0	216	...	False	
4	5	60	84.0	14260	8	5	2000	2000	350.0	655	...	False	

5 rows x 246 columns

Q9: Outlier Treatment:

a. Detect and handle any outliers in the dataset. Describe the methods used for detection and the rationale behind your chosen treatment strategy.

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files

- sample_data
- data_description_Feb2025.txt
- train_Feb2025.csv

```
[14] outlier_mask = (numeric_data < (Q1 - 1.5 * IQR)) | (numeric_data > (Q3 + 1.5 * IQR))

# Marking outliers as NaN for handling later, but only in the numerical columns
data[outlier_mask] = np.nan

display(data.isnull().sum()) # Checking missing values after marking outliers
```

	0
Id	0
MSSubClass	103
MSZoning	0
LotFrontage	348
LotArea	96
...	...
MoSold	0
YrSold	0
SaleType	0
SaleCondition	0
SalePrice	94

81 rows × 1 columns

dtype: int64

Q10: Missing Values Treatment: a. Identify and treat missing values in the dataset. Explain the techniques used for imputation or other treatments and justify your approach.

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files

- sample_data
- data_description_Feb2025.txt
- train_Feb2025.csv

```
[16] data[col].fillna(data[col].median(), inplace=True) # Filling numerical variables with median

<ipython-input-16-dfaacca36ecf>:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always
exists. For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or 'df[col] = df[col].method(va

data[col].fillna(data[col].mode()[0], inplace=True) # Filling categorical variables with mode (most frequent value)
```

	0
Id	0
MSSubClass	0
MSZoning	0
LotFrontage	0
LotArea	0
...	...
MoSold	0
YrSold	0
SaleType	0
SaleCondition	0
SalePrice	0

81 rows × 1 columns

dtype: int64

Q11: Scaling or Normalization: a. Apply suitable scaling or normalization techniques to the dataset. Describe the methods used and explain why they are appropriate for this data.

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Files

- sample_data
- data_description_Feb2025.txt
- processed_train_Feb2025.csv
- train_Feb2025.csv

```
[18] # Q11: Scaling or Normalization
# Standardizing numerical features using StandardScaler
scaler = StandardScaler()
numeric_cols = data.select_dtypes(include=[np.number]).columns # Selecting numerical columns
scaled_data = scaler.fit_transform(data[numeric_cols])
data[numeric_cols] = scaled_data # Updating dataset with scaled values

display(data.head()) # Display first few rows after scaling

# Save processed dataset to a new CSV file
data.to_csv("processed_train_Feb2025.csv", index=False)
```

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeat
0	-1.730865	0.412653	RL	-0.271614	-0.295059	Pave	Grvl	Reg	Lvl	AllPub	...	0.0	Gd	MnPrv	\$
1	-1.728492	-1.008276	RL	0.763241	0.103471	Pave	Grvl	Reg	Lvl	AllPub	...	0.0	Gd	MnPrv	\$
2	-1.726120	0.412653	RL	-0.064643	0.675276	Pave	Grvl	IR1	Lvl	AllPub	...	0.0	Gd	MnPrv	\$
3	-1.723747	0.767886	RL	-0.616566	0.086144	Pave	Grvl	IR1	Lvl	AllPub	...	0.0	Gd	MnPrv	\$
4	-1.721374	0.412653	RL	1.039202	1.718385	Pave	Grvl	IR1	Lvl	AllPub	...	0.0	Gd	MnPrv	\$

5 rows x 81 columns

Start coding or generate with AI.

processed_train_Feb2025.csv - Excel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	LotConfig	LandSlope	Neighborhood	Condition1	Condition2	BldgType	HouseStyle	OverallQual	OverallCond	YearBuilt	YearRemo	RoofStyle	RoofMatl	Ex
2	-1.73086	0.412653	RL	-0.27161	-0.29506	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	CollCr	Norm	Norm	1fam	2story	0.652641	-0.53427	1.060436	0.878668	Gable	CompShg	Vli
3	-1.72849	-1.00828	RL	0.763241	0.103471	Pave	Grvl	Reg	Lvl	AllPub	FR2	Gtl	Veenker	Feedr	Norm	1fam	1story	-0.07752	-0.53427	0.145062	-0.42958	Gable	CompShg	M
4	-1.72612	0.412653	RL	-0.06464	0.675276	Pave	Grvl	IR1	Lvl	AllPub	Inside	Gtl	CollCr	Norm	Norm	1fam	2story	0.652641	-0.53427	0.99263	0.830215	Gable	CompShg	W
5	-1.72375	0.767886	RL	-0.61657	0.086144	Pave	Grvl	IR1	Lvl	AllPub	Corner	Gtl	Crawfor	Norm	Norm	1fam	2story	0.652641	-0.53427	-1.923	-0.7203	Gable	CompShg	W
6	-1.72137	0.412653	RL	1.039202	1.718385	Pave	Grvl	IR1	Lvl	AllPub	FR2	Gtl	NoRidge	Norm	Norm	1fam	2story	1.382798	-0.53427	0.958727	0.733308	Gable	CompShg	W
7	-1.719	0.057421	RL	1.108193	1.668135	Pave	Grvl	IR1	Lvl	AllPub	Inside	Gtl	Mitchel	Norm	Norm	1fam	1.5fin	-0.80767	-0.53427	0.721408	0.49104	Gable	CompShg	Vli
8	-1.71663	-1.00828	RL	0.418289	0.271201	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	Somerst	Norm	Norm	1fam	1story	1.382798	-0.53427	1.094338	0.975575	Gable	CompShg	Vli
9	-1.71426	0.412653	RL	0.004347	0.374472	Pave	Grvl	IR1	Lvl	AllPub	Corner	Gtl	NWAmes	PosN	Norm	1fam	2story	0.652641	0.755038	0.043354	-0.57494	Gable	CompShg	Hc
10	-1.71188	0.057421	RM	-1.23748	-1.10252	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	OldTown	Artery	Norm	1fam	1.5unf	0.652641	-0.53427	-1.38056	-1.68937	Gable	CompShg	Br
11	-1.70951	0.057421	RL	-1.30647	-0.652	Pave	Grvl	Reg	Lvl	AllPub	Corner	Gtl	BrkSide	Artery	Artery	2fmCon	1.5unf	-0.80767	0.755038	-1.10934	-1.68937	Gable	CompShg	M
12	-1.70714	-1.00828	RL	0.073338	0.657948	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	Sawyer	Norm	Norm	1fam	1story	-0.80767	-0.53427	-0.22787	-0.96257	Hip	CompShg	Hc
13	-1.70477	0.412653	RL	1.108193	0.908849	Pave	Grvl	IR1	Lvl	AllPub	Inside	Gtl	Nridght	Norm	Norm	1fam	2story	2.112956	-0.53427	1.128241	1.024029	Hip	CompShg	W
14	-1.70239	-1.00828	RL	0.004347	1.270645	Pave	Grvl	IR2	Lvl	AllPub	Inside	Gtl	Sawyer	Norm	Norm	1fam	1story	-0.80767	0.755038	-0.32958	-1.10793	Hip	CompShg	Hc
15	-1.70002	-1.00828	RL	1.522135	0.466804	Pave	Grvl	IR1	Lvl	AllPub	Inside	Gtl	CollCr	Norm	Norm	1fam	1story	0.652641	-0.53427	1.182144	1.072482	Gable	CompShg	Vli
16	-1.69765	-1.00828	RL	0.004347	0.360915	Pave	Grvl	IR1	Lvl	AllPub	Corner	Gtl	NAmes	Norm	Norm	1fam	1story	-0.07752	-0.53427	-0.39738	-1.20483	Hip	CompShg	M
17	-1.69527	-0.1202	RM	-1.23748	-1.10252	Pave	Grvl	Reg	Lvl	AllPub	Corner	Gtl	BrkSide	Norm	Norm	1fam	1.5unf	0.652641	-0.53427	-1.44837	0.781761	Gable	CompShg	W
18	-1.6929	-1.00828	RL	0.004347	0.672157	Pave	Grvl	IR1	Lvl	AllPub	CulDSac	Gtl	NAmes	Norm	Norm	1fam	1story	-0.07752	2.044342	-0.05835	-0.7203	Gable	CompShg	W
19	-1.69053	1.478351	RL	0.211318	1.522135	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	SawyerW	RRAd	Norm	1fam	1story	-1.53783	-0.53427	-0.16006	-0.86566	Gable	CompShg	M
20	-1.68816	-1.00828	RL	-0.20262	1.522585	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	SawyerW	RRAd	Norm	1fam	1story	-0.80767	-0.53427	1.094338	0.972122	Gable	CompShg	Vli
21	-1.68578	-1.00828	RL	0.073338	-0.60349	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	NAmes	Norm	Norm	1fam	1story	-0.80767	0.755038	-0.46519	-0.96257	Hip	CompShg	Br
22	-1.68341	0.412653	RL	2.212038	1.70279	Pave	Grvl	IR1	Lvl	AllPub	Corner	Gtl	Nridght	Norm	Norm	1fam	2story	1.382798	-0.53427	1.128241	1.024029	Gable	CompShg	W
23	-1.68104	-0.1202	RM	-0.82354	-0.64195	Pave	Grvl	Reg	Bnk	AllPub	Inside	Gtl	IDOTRR	Norm	Norm	1fam	1.5unf	0.652641	2.044342	-1.41446	-1.68937	Gable	CompShg	Vli
24	-1.67867	-1.00828	RL	0.418289	0.152681	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	CollCr	Norm	Norm	1fam	1story	1.382798	-0.53427	1.026533	0.830215	Hip	CompShg	Vli
25	-1.67629	2.544048	RM	-1.72041	-1.75957	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	MeadowV	Norm	Norm	1fam	1story	-0.80767	2.044342	0.145062	-0.42958	Gable	CompShg	Ce
26	-1.67392	-1.00828	RL	0.004347	-0.36575	Pave	Grvl	IR1	Lvl	AllPub	Inside	Gtl	Sawyer	Norm	Norm	1fam	1story	-0.80767	-0.53427	-0.12616	0.781761	Gable	CompShg	Ph
27	-1.67155	-1.00828	RL	2.832951	1.707988	Pave	Grvl	Reg	Lvl	AllPub	Corner	Gtl	Nridght	Norm	Norm	1fam	1story	1.382798	-0.53427	1.196046	1.072482	Gable	CompShg	Vli
28	-1.66918	-1.00828	RL	-0.61657	-0.72824	Pave	Grvl	Reg	Lvl	AllPub	Corner	Gtl	NAmes	Norm	Norm	1fam	1story	-0.80767	2.044342	-0.70251	0.733308	Gable	CompShg	W
29	-1.6668	-1.00828	RL	2.005067	0.754288	Pave	Grvl	Reg	Lvl	AllPub	Inside	Gtl	Nridght	Norm	Norm	1fam	1story	1.382798	-0.53427	1.196046	1.120936	Gable	CompShg	Vli